

SatQuery AI — Geospatial Analysis Report

Remote Sensing Intelligence & Multimodal Evidence Verification

Job ID:	511f1880-392b-4d14-83ca-1959373780e9
Project ID:	e64343dc-4fb3-41c3-ac45-61635ae3beaf
Task Type:	UPERNET
Status:	RUNNING
Generated:	2026-09-26 11:38:40 UTC

1. Request Specification

Query / Prompt: "Segment scene into semantic land cover masks."

2. Factual Findings & Observations

Finding 1 (upernet_finding): Model UPerNet ConvNeXt-Tiny inference completed successfully.
Measurements: protocol_version: 1.0, success: True, job_id: f7d59384-5e3c-4d9a-aef3-f0c863dd7364, step_key: upernet_segmentation, attempt_id: 1, output_assets: {'segmentation_mask': 'C:\\Users\\HP\\earthquery\\storage\\scratch\\f7d59384-5e3c-4d9a-aef3-f0c863dd7364\\segmentation_mask.png'}, output_metadata: {'model': 'UPerNet ConvNeXt-Tiny', 'label_map': 'ADE20K Scene Parsing', 'detected_classes': [{'class_id': 0, 'name': 'wall', 'percentage': 44.13}, {'class_id': 21, 'name': 'water', 'percentage': 34.24}, {'class_id': 8, 'name': 'windowpane', 'percentage': 17.87}, {'class_id': 43, 'name': 'signboard', 'percentage': 2.97}, {'class_id': 1, 'name': 'building', 'percentage': 0.72}, {'class_id': 2, 'name': 'sky', 'percentage': 0.06}, {'class_id': 26, 'name': 'sea', 'percentage': 0.01}], 'note': 'Pretrained ADE20K classes represent demo proxy. Satellite land cover requires domain-specific head.'}, metrics: {'duration_ms': 733.1128120422363, 'peak_vram_mb': 1611.1220703125, 'peak_ram_mb': None, 'device_name': 'cuda', 'cudnn_enabled': False}, error_code: None, error_message: None

3. Operational Boundaries & Limitations

- Observations are generated based on local hardware inference (RTX 5060 Laptop GPU).
- ChangeFormer outputs represent 2D building change masks trained on LEVIR-CD.
- UPerNet semantic segmentation reflects generic ADE20K demo proxy classes unless domain-trained.
- Satellite pixel coordinates are not mapped to geographic bounds unless explicit CRS metadata is verified.
- Analyst confirmation is recommended prior to operational deployment.